

Track 38: MATLAB & Simulink for Engineers

June 2026

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Departments: EEE · ECE · Mechanical · AI & DS · AI & ML | MR ROKESH TECHNOLOGY

Skill path: MATLAB Scripting → Matrix Ops → Plotting → Simulink → Control Systems → Signal Processing → Simulation Projects

Related: [Department Map](#) | [Track 39 — Power Systems](#)

Track Overview

Who this is for: Engineering students across EEE, ECE, Mechanical, and AI branches needing simulation and numerical computing skills.

Prerequisites: Basic calculus and physics; programming helpful but taught from basics.

Tools: MATLAB, Simulink, Control System Toolbox, Signal Processing Toolbox, student license.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	MATLAB literacy	10 scripts + plots
1 Month	80–100	Simulation basics	Control system Simulink model
3 Months	250–300	Project-ready	Domain-specific simulation capstone
6 Months	500–600	Engineer-ready	Multi-domain simulation portfolio

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Variables, matrices, functions, plotting	5 MATLAB scripts
D3–4	Symbolic math, ODE solvers, curve fitting	Data analysis report
D5–6	Simulink basics: blocks, transfer functions	First Simulink model
D7–8	PID control, step response, tuning	Motor control simulation
D9–10	Capstone	Simulink project + report

1-Month Foundation

Week	Topics	Project
W1	MATLAB programming, file I/O, debugging	Numerical methods toolkit
W2	Simulink: subsystems, S-functions intro	Multi-block simulation
W3	Control systems: Bode, root locus, stability	Feedback control design
W4	Signal processing: FFT, filtering	Signal analysis project

3-Month Job Ready (Pick Domain)

Domain	Month 3 Capstone
EEE	Power system load flow simulation
ECE	Digital filter design + implementation
Mechanical	Vibrational analysis of mechanical system
AI/ML	ML algorithm implementation in MATLAB

6-Month Professional

Full control systems + state-space design + Simscape physical modeling + 3 domain projects + MATLAB certification prep.

Assessment Rubric

Weekly Quiz 20% | Script/simulation labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Assign domain-specific capstone based on student's department. MathWorks Onramp courses as Week 1 homework.